

MUSKAN MITTAL

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SUMMARY

Data Scientist with 2+ years of production experience building end-to-end ML pipelines — data ingestion, model training, and cloud-deployed inference APIs. Delivered a medical imaging CNN achieving 88% accuracy on Azure, and engineered ML data systems for a US life-sciences firm. Proficient in TensorFlow, deep learning, NLP, GenAI integration, and REST API development.

EXPERIENCE

- Data Scientist | Mlytics Life Sciences — US (Remote)**

Jul 2024 – Present

 - Architected Python + SQL ML data pipelines, cutting model training data-prep time by ~40% and accelerating team iteration cycles across large-scale life-sciences datasets.
 - Automated 5+ reporting and validation workflows, reducing analyst effort ~60% per sprint; owned ML data infrastructure enabling 3 models to reach production on schedule.
 - Engineered scalable backend data systems for real-time ML workflows, eliminating pipeline failures under peak load and improving data quality standards across the organization.
 - Collaborated with cross-functional teams to translate business requirements into ML-ready data solutions, ensuring alignment between data pipelines and model performance objectives.
- Data Scientist Intern | Mlytics Life Sciences — US (Remote)**

Feb 2024 – May 2024

 - Optimized SQL pipelines for ML training jobs, cutting data retrieval latency ~35%; built inference dashboards giving stakeholders real-time model performance visibility for the first time.
 - Resolved 10+ backend data inconsistencies that had been silently degrading model prediction quality, and contributed to reporting systems for ML-driven decision-making.
 - Assisted in feature engineering and data preprocessing for model training datasets, applying transformation and cleaning techniques that improved downstream model accuracy.

PROJECTS

- Skin Cancer Detection System** | TensorFlow · CNN · Flask · Microsoft Azure · Streamlit

 - Built a CNN image classifier using TensorFlow/Keras achieving 88% diagnostic accuracy on dermoscopic images; applied k-fold cross-validation, dropout regularization, and data augmentation for clinical robustness.
 - Exposed a Flask-based RESTful API serving model inference with sub-2s latency; deployed complete end-to-end pipeline on Microsoft Azure App Services as a publicly accessible production web application.
 - Implemented multi-class classification across 7 skin lesion categories, enabling the model to distinguish between benign and malignant conditions with clinically meaningful precision and recall scores.

Live Demo: [skin-streamlit-app2-cahtfd3epa4d9fp.centralindia-01.azurewebsites.net](#)
GitHub: github.com/Muskan29mm/Skin-Cancer-Detection-System-using-CNN-
- CNN Text Classifier** | TensorFlow · Keras · spaCy · NLP · Streamlit

 - Developed a multi-class CNN text classifier using TensorFlow/Keras with embedding layers and conv filters; integrated a spaCy NLP preprocessing pipeline — tokenization, lemmatization, NER, and custom text normalization.
 - Built modular backend separating preprocessing, inference, and API logic; deployed as a real-time interactive Streamlit app enabling live multi-class predictions via public URL.
 - Evaluated model performance using precision, recall, F1-score, and confusion matrix analysis; iteratively tuned hyperparameters and embedding dimensions to optimize classification accuracy across all categories.

Live Demo: [cnn-text-classifier-nhxzqdbvkzrqzpy5amuxx.streamlit.app](#)
GitHub: github.com/Muskan29mm/CNN-text-Classifier

SKILLS

Programming:	Python, SQL, Django
ML / DL:	Scikit-learn, TensorFlow, PyTorch, Supervised & Unsupervised Learning, Model Training, Hyperparameter Tuning, Feature Engineering, Model Evaluation
NLP:	spaCy (NER, POS Tagging, Text Preprocessing, Custom Pipelines), Text Classification, Information Extraction
Generative AI:	LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Chatbot Development
Cloud / MLOps:	Microsoft Azure, AWS (Basics), Model Deployment, REST APIs (Flask/FastAPI)
Data Engg:	Pandas, NumPy, Data Cleaning, Data Transformation, ETL Pipelines
CV:	OpenCV, Image Classification, Object Detection, Image Processing
Tools:	Git/GitHub, PostgreSQL, MySQL, OOP, Data Structures & Algorithms

EDUCATION

- M.C.A. — Thapar Institute of Engineering & Technology, Patiala, Punjab2022 – 2024
- B.Sc.IT — RIMT University, Mandi Gobindgarh, Punjab2019 – 2022